Agenda:

1️⃣ Learn Core DevOps Tools (Focus on Enterprise-Relevant)

Target tools and practices that show you can **build, deploy, and monitor real systems**:

Skill / Tool	Why Important	Practical Exercise
Docker / Kubernetes	Containerization and orchestration	Spin up a kind cluster locally and deploy nginx + multi-service apps
CI/CD (Jenkins, GitHub Actions, GitLab CI)	Automating deployments	Create a pipeline that builds, tests, and deploys a container
IaC (Terraform, Ansible)	Automate infrastructure	Write Terraform to deploy a VM + network, Ansible to configure it
Monitoring (Prometheus, Grafana, ELK)	Enterprise monitoring	Deploy Prometheus in Kubernetes, track pod metrics
Git / GitHub	Version control	Keep all projects on GitHub for visibility

2️⃣ Build a Portfolio (Highly Recommended)

Tips:

- **Start small:** One or two projects showing core DevOps principles
- **Show end-to-end workflow:** Code → Build → Test → Deploy → Monitor
- **Document your work:** README, architecture diagrams, screenshots
- **Keep it accessible:** GitHub, GitLab, or even a personal website
- **Use real enterprise scenarios:** Example: "Deploy multi-node Kubernetes cluster, expose NGINX, add monitoring, implement IAM policies"

Suggested First Projects

1. **Local Kubernetes lab**
    - Spin up kind cluster, deploy nginx and a sample app
    - Show port-forward, NodePort, services, and basic networking

2. **CI/CD pipeline demo**
    - Build a small Python or Node.js app
    - Push to GitHub → auto-build → deploy to Kubernetes cluster
3. **Infrastructure as Code lab**
    - Use Terraform to provision VMs or Docker hosts in local environment or cloud sandbox
    - Configure with Ansible
4. **Monitoring & alerting demo**
    - Prometheus + Grafana on Kubernetes
    - Visualize pod metrics or custom metrics

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3 Make It Public and Trackable

- GitHub is the **standard**
- Add screenshots, architecture diagrams, maybe even a short demo video
- Consider using **free cloud sandboxes** for projects (like AWS free tier, GCP, or Azure sandbox)

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4 Networking and Visibility

- Engage in **LinkedIn posts** about your projects and learnings
- Join DevOps communities (Slack, Discord, Reddit)
- Contribute small scripts or modules (Ansible roles, Helm charts, Terraform modules) — this is portfolio material too

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5 Mindset: Enterprise DevOps

For enterprise employers:

- Security, compliance, and reliability matter **more than cutting-edge tech**
- IAM + storage + networking expertise makes a strong candidate if can **connect it to automation and DevOps workflows**

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6 Timeline (Efficient Approach)

- **Month 1–2:** Docker + Kubernetes basics (kind, port-forward, simple pods)
- **Month 2–3:** CI/CD pipelines + GitHub Actions
- **Month 3–4:** Terraform + Ansible + small cloud lab
- **Month 4–5:** Monitoring (Prometheus/Grafana) + logging
- **Month 5+:** Build portfolio, post projects, start applying

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